6.3 Ecosystems

6.3.1 Ecosystems

Organisms do not live in isolation but engage in complex interactions, not just with other organisms but also with their environment.

The efficiency of biomass transfer limits the number of organisms that can exist in a particular ecosystem.

Ecosystems are dynamic and tend towards some form of climax community.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	ecosystems, which range in size, are dynamic and are influenced by both biotic and abiotic factors	To include reference to a variety of ecosystems of different sizes (e.g. a rock pool, a playing field, a large tree) and named examples of biotic and abiotic factors.

(b) biomass transfers through ecosystems	To include how biomass transfers between trophic levels can be measured **AND** the efficiency of biomass transfers between trophic levels $efficiency = \frac{biomass\ transferred}{biomass\ intake} \times 100$ **AND** how human activities can manipulate the transfer of biomass through ecosystems. *M0.1, M0.2, M0.3, M0.4, M1.1, M1.3, M1.6* HSW12
(c) recycling within ecosystems	To include the role of decomposers and the roles of microorganisms in recycling nitrogen within ecosystems (including *Nitrosomonas*, *Nitrobacter*, *Azotobacter* and *Rhizobium*) **AND** the importance of the carbon cycle to include the role of organisms (decomposition, respiration and photosynthesis) and physical and chemical effects in the cycling of carbon within ecosystems. HSW2, HSW12
(d) the process of primary succession in the development of an ecosystem	To include succession from pioneer species to a climax community **AND** deflected succession. HSW12
(e) **(i)** how the distribution and abundance of organisms in an ecosystem can be measured **(ii)** the use of sampling and recording methods to determine the distribution and abundance of organisms in a variety of ecosystems.	*M1.3, M1.4, M1.5, M1.7, M1.9, M1.10, M3.1, M3.2* **PAG3** HSW4